

OM M. PATEL

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Vector AI Scholar and Applied AI engineer building production AI Integrations, data pipelines, and backend systems to help teams make better operational decisions. Experience spans across evaluating AI, agent routing loops for human review, LLMs fine-tuning, and NLP research.

EDUCATION

Master of Science in Computer Science, with AI Specialization (GPA: 3.77/4)

York University, Ontario, Canada

Sept 2025 – August 2026

Relevant Courses: Fairness and Bias in AI, Robotics, Data Mining, Probabilistic Models and Machine Learning, Machine Learning Theory, Data Analysis and Visualization.

SKILLS

Languages	TypeScript, Python, Java, SQL
ML/NLP/LLM	PyTorch, Pandas, Scikit-learn, LangChain, NumPy, QLoRA, OpenAI SDK, Google ADK.
Backend & Data	React, FastAPI, Flask, Next.js, Airflow, PostgreSQL, Supabase, Neo4j, ChromaDB
Tools	Docker, CI/CD, GitHub, Jira, Vercel, GCP, Cloudflare, Postman

INDUSTRY EXPERIENCE

Applied AI Engineer (Co-op)

June 2026 – August 2026

BarLens, Toronto

- Integrated **Gemini vision API** with **Next.js** to read labels and reduced manual work by **64%**.
- Streamlined Human-verified captures by **HITL validation**, POS reconciliation, and dashboards.
- Designed **Supabase**-backed data storage, private media retention, and audit-ready ledgers.
- Deployed on **Cloudflare**, running live on **20+ locations**.
- Reviewed **AI Agents** to deliver skills for: automated tests, **CI** checks, and sandbox QA to support reliable releases.

Data Engineering Intern

Dec 2024 – Aug 2025

Uarra, Nigeria(Remote)

- Built Airflow-powered **ETL pipelines** to parse geospatial datasets using GeoPandas to increase data throughput by around **23%**.
- Improved **query performance** by optimizing **PostgreSQL** schemas and indexing for real-time lookups used by downstream ML and analytics components.
- Coordinated with engineering and ML teams** to monitor API behavior and support iterative post-deployment improvements to data and model-backed APIs.

PROJECTS

Synapse AI : Teacher Orchestrated Student Learning Platform

May 2026

- Won a Hackathon building AI agent led teaching platform supporting diagnostics, tutor streaming, classroom invites, and teacher analytics.
- Built with **FastAPI** **Next.js**, **ElevenLabs API** to transform uploaded learning materials into personalized study outputs such as notes, flashcards, quizzes, and podcasts.

TTC Interactive Dashboard : Service Dynamics & Delays

Jan 2026 – April 2026

- Co-developed a **Python ingestion and normalization pipeline** joining TTC delay data with static GTFS and live service alerts.
- Processed over **1 million delay events** into **DuckDB** and Parquet data marts, with checks for unmatched route and station mappings and reproducible Streamlit views.
- Integrated **Google ADK** for in-app AI bot support to dive deeper into analysis for better UX.

Legal Clarity : Simplifying Legal Documents

Aug 2024 – Nov 2024

- Fine-tuned transformer models (**Pegasus**, **T5**, **IndicBART**) in **PyTorch** to summarize and translate long-form, multilingual legal text for faster review.
- Packaged the pipeline into a **Flask** service for repeatable inference and downstream integration.

Driver Pay Forecasting : Uber NYC Driver Pay Prediction Compared ML-DL models, ran architecture and hyperparameter searches achieving almost 12% lower RMSE vs. baseline. Visualized features affecting driver pay on Tableau.	Feb 2024 – May 2024
ChatPdf AI : Chat with Your PDF - Built a retrieval-first document QA system using LangChain and FAISS to ground answers from source text. - Integrated Agents SDK for agentic workflows in document querying and response generation, resulting in 80% accuracy on manual test set($N \leq 50$). Deployed on Streamlit.	Dec 2023 – Feb 2024
Autograder : Precision in Every Grade Led a 6-person team to build an auto-grading pipeline (BERT-uncased, SpaCy, PyTesseract) with React front-end and Flask backend, enabling semi-automated grading for handwritten and typed responses	Sept - Nov 2023

RESEARCH & PUBLICATIONS

Sentiment Detection and Cross-Lingual Preservation <i>Research Assistant under Prof. Uyen T. Nguyen (York University)</i> <i>Manuscript in preparation</i> - Built a Gujarati sentiment-analysis pipeline across translated SST-2 and native GSAC data to evaluate polarity preservation in English–Gujarati transfer. - Benchmarked multilingual and Indic-focused NLP models, including TF-IDF baselines, and MuRIL. - Designed disjoint GSAC adaptation/test evaluation and analyzed polarity shifts, negation errors, mistranslations, and social-text noise for the manuscript.	March 2026 – Present
LLMs for Optimization : Job Shop Scheduling Optimization (JSSP) <i>Technical Project under Prof. Aijun An (York University)</i> - Built a GNN-augmented LLM pipeline for scheduling by converting JSSP instances into disjunctive graphs (PyTorch Geometric) and serializing constraints into structured prompts. - Pre-processed the noisy STARJOB dataset (~130k to 9,525 usable instances), reducing fine-tuning runtime from 70h to 11.5h using 4-bit LoRA (Unsloth) on an NVIDIA RTX A6000.	Sept 2025 – Dec 2025
ADCIS 2024 (Springer): Automated Sleep Stage Classification using Machine Intelligence Techniques <i>Co-author</i> - Surveyed 90+ automated sleep staging studies and organized signal modalities (EEG/ECG/EOG/EMG) and PSG data representations into an end-to-end taxonomy used for system design. - Consolidated model families and evaluation practice (classical ML → deep learning) with reporting guidance around leakage, imbalance, and cross-subject generalization.	Sept 2024 – Jul 2025

AWARDS & LEADERSHIP

Vector AI Scholarship Recipient, York University	May 2025
Top-10 rank Indian weather prediction at Kaggle ML Olympiad	April 2024 – May 2024
AI/ML Mentor: Mentored 10+ students on projects including Autograder and Legal Clarity	